Decision Assistance and Steering in Health Insurance
SUPPLEMENTAL APPENDIX

Ian McCarthy & Evan Saltzman

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1 Constructing the Outside Option

In this study, we model dynamic switching both between plans and into and out of the exchange market (any plan, vs. remaining uninsured) using five years of longitudinal data on exchange customers. Previous studies of the exchanges have treated demand as static, typically merging administrative data on exchange enrollees with survey data such as the American Community Survey (ACS) on the uninsured to form the universe of potential exchange consumers (Tebaldi, 2025; Saltzman, 2019). In contrast, we form the uninsured population using data on consumers in our Covered California data for years in which they did not have exchange coverage, and using panel survey data on dynamics of consumers' exchange market eligibility. For example, within our data, consumers with exchange coverage in 2015 and 2016 are considered uninsured (the outside option) in 2014, 2017, and 2018, provided they remained eligible for the exchange market. Consumers might lose exchange market eligibility if they gain access to employer-sponsored insurance or public insurance such as Medicaid or Medicare. Because our data do not indicate when enrollees lose exchange market eligibility, we use the Survey of Income and Program Participation (SIPP) to impute consumer eligibility.

To impute consumer exchange market eligibility, we use data from the SIPP. The SIPP is well-suited for the imputation because (1) it asks the insurance status of respondents for every month over a three-year period (2013-2015) and (2) it includes detailed information on the chief reasons for consumers' coverage status, such as whether the respondent obtained or lost an offer of employer-sponsored insurance, moved in or out of California, or became eligible or ineligible for Medicare or Medicaid. For SIPP respondents who newly obtained or gave up individual market coverage, we construct a *transitioned* variable that indicates whether the respondent gained or lost eligibility for the individual market. The *transitioned* variable takes value 1 if the respondent (1) belongs to a household that obtained or lost an offer of employer-sponsored insurance; (2) moved out of or into California; (3) the respondent turned 65 and qualified for Medicare; and (4) the respondent became eligible for Medicaid following a drop in income. We estimate a logit model regression of the *transitioned* variable on the demographic variables available in both the SIPP and the Covered California data, including age, income, gender, race, and household size. We use the estimated logit to predict whether Covered California consumers observed for only some years of our study timeframe transitioned into or out of the exchange market. Consumers transitioning into or out of the exchange market are removed from our study population during years when they are not enrolled in an exchange plan.

2 Additional Reduced-Form Results

The reduced-form estimates in the body of the paper come from a prediction-based estimator. We fit the outcome model on unassisted households, weighted by inverse propensity weights, predict the counterfactual outcome for assisted households, and form the ATT as the difference between what assisted households chose and what the model predicts they would have chosen. Assistance never appears as a regressor, so there is no assistance coefficient to report and no control function anywhere in the estimation. This appendix reports the complementary pooled specifications, in which assistance does enter as a covariate and its coefficient can be traced across a progressive sequence of control sets.

2.1 Dominated Choices

Table A1 reports linear probability models of the dominated-choice indicator on the sample of households with at least one dominated plan in the choice set. Column (1) is unconditional. Column (2) adds the household demographics that enter the propensity score, namely age composition, gender, race and ethnicity, and household size. Income does not appear, because the sample is defined by cost-sharing-reduction eligibility and therefore contains only households below 200% of the federal poverty level, leaving the income brackets used elsewhere in the paper with no variation to exploit. Column (3) adds a new-enrollee indicator and year fixed effects. Columns (4) and (5) are the two terminal specifications. Column (4) adds the broker-density control function without region fixed effects, and column (5) adds region fixed effects without the control function. Standard errors are clustered by rating region throughout, and all specifications are inverse-propensity weighted.

The assistance coefficient is stable across columns (1) through (3) and (5), moving only from -0.0152 to -0.0168 as controls accumulate and settling at -0.0161 once region fixed effects absorb differences in market composition. Column (4) is the exception. With the control function included, the assistance coefficient falls to -0.2564 and the control function residual enters at $+0.2402$, so that the two nearly offset and their sum, -0.0162, reproduces the estimate in column (3). This is what a first stage too weak to separate selection from the treatment effect produces, and it is a reparameterization rather than independent evidence.

The two terminal columns never appear together, and the reason is not that the combination would merely be underpowered. Broker density varies only across rating region and year, so region fixed effects leave the instrument with no variation to exploit. Within a region the number of licensed agents is close to a deterministic function of the year, which means a region-specific control function is identically zero rather than weak. Clustering standard errors at the level at which the instrument actually varies, the partial F statistic on broker density is 3.6 and the incremental R^2 is 0.001, against a partial F of roughly 7,600 when the clustering is ignored. The apparent strength of the instrument in the unclustered specification reflects the number of households rather than the number of markets. The instrument itself is a sensible one, and its first-stage relationship with assistance take-up is positive and in the expected direction, but it rests on too few markets for meaningful inference. Nonetheless, even with a weak first stage, the pooled results are consistent with the body estimates. The implied total effect in column (4) matches the stable columns, and every specification in the table points to the same modest reduction in dominated choices.

Table A2 reports the body estimator re-run on new enrollees alongside the all-enrollee estimates. The body sample is all enrollees so that the reduced form and the structural model are estimated on the same households. Restricting to new enrollees produces larger effects for every channel, consistent with assistance mattering most for households making a first plan choice, and leaves the ordering across channels unchanged.

Table A1: Dominated Choices, Pooled Specifications[a]

Variable	(1)	(2)	(3)	(4)	(5)
Assisted	-0.0152***	-0.0152***	-0.0168***	-0.2564**	-0.0161***
	(0.0012)	(0.0011)	(0.0011)	(0.1181)	(0.0007)
CF Residual				0.2402*	
				(0.1185)	
HH Size		-0.0018***	-0.0007	0.0120*	-0.0003
		(0.0004)	(0.0005)	(0.0064)	(0.0005)
New Enrollee			0.0168***	0.0167***	0.0168***
			(0.0027)	(0.0027)	(0.0027)
Demographics		X	X	X	X
Year FE			X	X	X
Region FE					X
Control Function				X	
Observations	2,993,294	2,993,294	2,993,294	2,993,294	2,993,294
R^2	0.002	0.003	0.012	0.013	0.016

[a]Linear probability models of an indicator for selecting a dominated plan, on the sample of households with at least one dominated plan in the choice set. All specifications are weighted by inverse propensity weights, with standard errors clustered by rating region. Demographics include age composition, gender, race and ethnicity, and household size. Columns (4) and (5) are the two terminal specifications described in the text and are never combined. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A2: Dominated Choice ATT, All Enrollees Versus New Enrollees[a]

Channel	All Enrollees	New Enrollees
Any Assistance	-0.0156	-0.0226
Agent/Broker	-0.0183	-0.0252
Navigator	-0.0090	-0.0180

[a]Prediction-based ATT of decision assistance on the probability of selecting a dominated plan, weighted by inverse propensity weights. The all-enrollee column is the body estimate. The new-enrollee column re-runs the same estimator on households enrolling for the first time.

2.2 Plan Choice

Table A3 reports the companion sequence for plan choice. The model is a conditional logit over the plans in each household's choice set, pooled across assisted and unassisted households and weighted by inverse propensity weights. Assistance enters through its interaction with metal tier, separately by channel, using the same navigator and broker split as the structural model in Section 5. Column (1) includes plan attributes only. Column (2) adds the premium-by-demographic interactions that constitute the full control set in the body specification. Column (3) adds the broker-density control function, again interacted with plan attributes.

Three features of this model differ from the dominated-choice table and are worth stating rather than leaving to inference. First, there is no region fixed effects column. A household-level region indicator is constant across the alternatives in that household's choice set and is differenced out of the conditional logit, so it does not control for anything. The term that would actually absorb region here is region interacted with metal tier and insurer, which adds well over a hundred parameters and leaves the insurer block empty wherever an insurer does not operate in a region. We do not estimate it, and region is therefore uncontrolled in this model rather than absorbed.

Second, assistance interacts only with silver and bronze. Catastrophic plans are excluded from the sample, so the four metal tiers exhaust the alternatives and a full set of four assistance-by-metal interactions sums to a household constant, which is differenced out. Gold and platinum share the omitted category, matching the base specification.

Third, the control function terms enter as interactions with plan attributes for the same reason. The first-stage residual is a household-level quantity and cannot enter a conditional logit on its own.

Column (3) shows the same pattern as the control-function column of the dominated-choice table. The assistance-by-silver interactions inflate and the control-function-by-silver term moves in the opposite direction, consistent with the weak first stage documented above, and we read the column as a reparameterization of column (2) rather than independent evidence. The steering pattern itself, toward silver and away from bronze with brokers steering more strongly than navigators, is stable across all three columns.

Table A3: Plan Choice, Pooled Specifications[a]

Variable	(1)	(2)	(3)
Premium	-0.9563 (0.0021)	-0.5257 (0.0043)	-0.4857 (0.0044)
Navigator × Silver	0.2960 (0.0093)	0.2829 (0.0094)	1.8499 (0.0268)
Navigator × Bronze	-0.3046 (0.0104)	-0.3311 (0.0105)	-0.3471 (0.0343)
Broker × Silver	0.6039 (0.0077)	0.6007 (0.0078)	2.0867 (0.0252)
Broker × Bronze	0.0562 (0.0084)	0.0581 (0.0086)	0.0239 (0.0328)
CF × Silver			-1.5822 (0.0263)
CF × Bronze			0.0008 (0.0339)
Premium × demographics		X	X
Control function			X
Households	1,067,827	1,067,827	1,067,827
Log-likelihood	-2,668,062	-2,619,902	-2,608,811

[a]Conditional logit over the plans in each household's choice set, pooled across assisted and unassisted households and weighted by inverse propensity weights. Premiums are net of subsidy and measured in hundreds of dollars. Plan attributes are metal tier, network type, health savings account eligibility, and insurer indicators. CF denotes the broker-density control function interacted with the indicated plan attribute. Standard errors in parentheses.

References

Saltzman, Evan, "Demand for health insurance: Evidence from the California and Washington ACA exchanges," *Journal of health economics*, 2019, *63*, 197–222.

Tebaldi, Pietro, "Estimating Equilibrium in Health Insurance Exchanges: Price Competition and Subsidy Design under the ACA," *The Review of economic studies*, 2025, *92* (1), 586–620.